

EE 541: Tensors: Techniques, Algorithms, Applications for Signal Processing, and Machine Learning.

Course Code	EE 541
Course Name	Tensors: Techniques, Algorithms, Applications for Signal Processing, and Machine Learning.
L-T-P-C	3-0-2-4
Prerequisites	Linear algebra, basic probability, and statistics.
Intended for	B.Tech. (3rd/4th year)/ MTech/ MS/ PhD of the SCEE.
Mutual Exclusion	None (at present)
Approval	50th BoA (Jul 2023)

Course Contents

- **Linear algebra recap:** Vector spaces, subspaces, linear in/dependence, bases, dimensions, principle of orthogonality and projections, linear models least-squares problems, Rank-decomposition for matrix, SVD and low-rank matrix approximation. (5 hours)
- **Working with Tensors:** Useful products and their properties (Inner, Outer, Hadamard, Kronecker, and Khatri-Rao, and mode-n). Tensor, its different views, and reshaping. Operators on tensors, tensor contraction, their algebraic properties, tensor rank, low rank tensor approximation, Tensor calculus and its fundamental operations. (10 hours)
- **Tensor factorization and its computation:** Rank decomposition for tensor, CP decomposition (CANDECOMP/ PARAFAC), properties of CP decomposition, Hardness of CP decomposition, algorithms for computing decompositions (ALS, Jennrich's algorithm, etc.); Other notions of tensor decomposition: Tucker decomposition, HOSVD (higher order SVD), Tensor train decomposition (TT-SVD). (9 hours)
- **Techniques for compressing tensors:** Dimensionality reduction (random projection) for vectors, Tensorized random projection, Compressing Tensors using Count Sketch, Higher Order Count Sketch. (8 hours)
- **Applications:** Blind Multiuser CDMA, Blind Source Separation, Harmonics, Gaussian Mixture parameter estimation, learning latent variables, Topic modelling, Learning Hidden Markov Models, Community detection, Collaborative filtering-based recommender systems, including recent ML/ SP based approaches. (10 hours)

Laboratory/ practical/ tutorial Modules: The two-hours of lab session will enhance the understanding of the concepts taught in the class. The lab will cover the concepts including, principle of orthogonality, least Squares, SVD, low-rank matrix decomposition, Inner, Outer, Hadamard, Kronecker, and Khatri-Rao products, mode-n Tensor, PARAFAC, HOSVD, TT-SVD, ALS, low-rank tensor decomposition, etc.

Text Books

1. J. Landsberg, **Tensors: Geometry and Applications**, vol. 128. Providence, RI, USA: Amer. Math. Soc., 2011.
2. Haiping Lu, Konstantinos N. Plataniotis, Anastasios Venetsanopoulos, **Multilinear Subspace Learning - Dimensionality Reduction of Multidimensional Data**, 1st Edition, CRC press, 2015.

References

1. Ankur Moitra, **Algorithmic aspects of machine learning**, Cambridge University Press, 2018.
2. T. G. Kolda, B. W. Bader, **Tensor Decomposition and Applications**, SIAM Review, 2009.
3. N. D. Sidiropoulos, L. De Lathauwer, X. Fu, K. Huang, E. E. Papalexakis, and C. Faloutsos, **Tensor decomposition for signal processing and machine learning**, IEEE Trans. Signal Process., vol. 65, no. 13, pp. 3551–3582, Jul. 2017.